

TASHKENT INTERNATIONAL UNIVERSITY · FACULTY OF ECONOMICS Continuous
distributions and normal distribution

Week 5 · Lectures 1 and 2

Tashkent International University · Faculty of Economics

2025/2026 academic
year



From Sum to Integral

Week 4. $\square$ was counted: 0, 1, 2, ... Each value had its own probability:

$$E[\square] = \sum_x x p(x)$$

There are no such values. The probability of each specific value is zero.

Discrete $\square(\square) = P(\square = \square)$

Continuous $\square(\square)$ — density

Probability =
area under the
curve

$\Sigma \rightarrow \int$

This week the definite integral in MATH102 comes into play. The idea doesn't change—only the sum becomes an integral.

Learning Objectives for Week 5

By the end of this week, you will:

1. Define the density function and explain why $P(X = x) = 0$.
2. Calculate the probability as a surface; show the relationship between μ and σ .
3. Work with uniform and exponential distributions.
4. Standardize the normal distribution and use the z -estimate.
5. Assess normality using a Q-Q plot.
6. Approximate the binomial to normal and apply the continuity correction.

Learning Outcome CLO3. Working with Random Variables and Standard Distributions (Bloom: Application and Analysis).

Common Mistakes

Math Requirement. Definite Integral $\int_a^b f(x) dx$ — MATH102, Chapter 5. If you forget, review before class.

Density function

Density Function

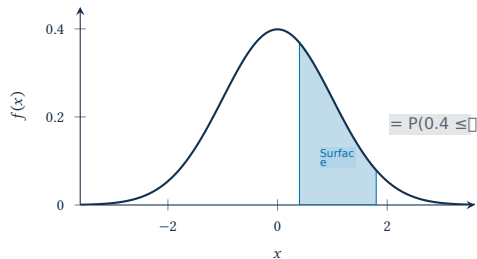
Definition Density Function $f(x)$:

$$P(a \leq x \leq b) = \int_a^b f(x) dx$$

Conditions:

$$f(x) \geq 0, \quad \int_{-\infty}^{\infty} f(x) dx = 1$$

A common misconception is that $f(x)$ is not a probability. It can be greater than 1. Probability is just a surface.



$$P(x = x) = \int_x^x f(t) dt = 0$$

Therefore, in the continuous case, $P(x < x) = P(x \leq x)$.

Uniform distribution: the simplest continuous model

$$f(x) = \frac{1}{b-a}, \quad a \leq x \leq b$$

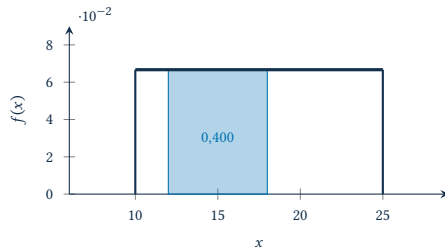
$$E[X] = \frac{a+b}{2}, \quad \text{Var}(X) = \frac{(b-a)^2}{12}$$

Example. $X \sim \text{Unif}(10, 25)$:

$$f(x) = \frac{1}{15} = 0,0667$$

$$E[X] = 17,5, \quad \text{Var} = 18,75, \quad \text{SD} = 4,33$$

$$P(12 < X < 18) = \frac{18-12}{15} = 0,400$$



Audience Question What is $P(X = 15)$?
How does it look on the graph?

Normal distribution

Normal distribution

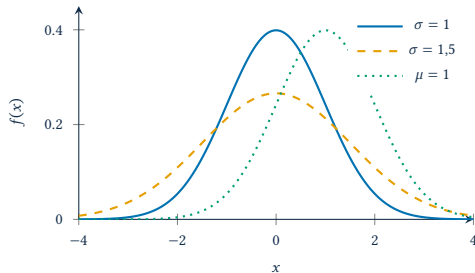
Definition
 $X \sim N(\mu, \sigma^2)$:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

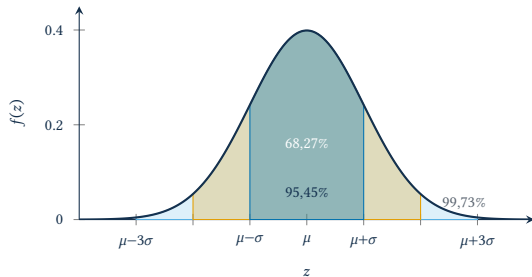
μ — center, σ — dispersion.

Properties:

- ▶ Symmetrical around μ
- ▶ Bell-shaped, single-peaked
- ▶ μ = median = mode
- ▶ Determined on the integer axis



68-95-99.7 rule



Interval Exact probability	
$\mu \pm 1\sigma$	0,682689
$\mu \pm 2\sigma$	0,954500
$\mu \pm 3\sigma$	0,997300

Common mistake The opposite direction is also needed: for 95% certainty, $\alpha = 1.960$, not 2. For 99% certainty, $\alpha = 2.576$.

These numbers appear regularly in confidence intervals starting in week 9.

Standardization: z-score

Definition Standard score (z-score)

$$Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1)$$

z — shows how many standard deviations away the observation is from the mean.

Why is it needed? One table (or one function) is enough for all normal distributions.

$\Phi(z) = P(Z \leq z)$

−1,960	0,0250
−1,645	0,0500
−1,000	0,1587
0,000	0,5000
1,000	0,8413
1,645	0,9500
1,960	0,9750
2,576	0,9950

A common mistake is that z is unitless. Therefore, soums, tons, and points are compared on the same scale.

Normality check

Same variable, two countries, two different answers

Skewness -0.649 ; Q-Q $\hat{\rho} = 0.9405$;
Shapiro- Wilk $\hat{\rho} = 0.866$, $\hat{\rho} = 0.0025$.

Reason: distribution is bimodal (week 1).

Close to normal: Kyrgyzstan

$n = 27$; $\bar{x} = 2,235$, $s = 0,277$.

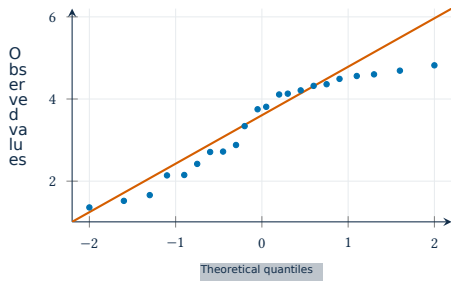
Skewness -0.655 ; Q-Q $\hat{\rho} = 0.9660$;
Shapiro- Wilk $\hat{\rho} = 0.929$, $\hat{\rho} = 0.0640$.

The normal model is valid here.

Audience Question Both have almost the same skewness (-0.65), but one is normal and the other is not. Which indicator showed the difference?

One number is not enough — look at the graph.

Q-Q plot: visual test for normality



Uzbekistan: points systematically deviate from the line

How to read:

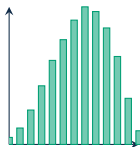
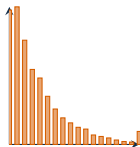
- Points on the line $\Rightarrow$ normal
- $\cap$ -shaped $\Rightarrow$ skewness
- Rising ends $\Rightarrow$ heavy tails
- Stairs $\Rightarrow$ discreteness or groups

Common mistake Be careful with small n . At $n = 27$, the points naturally “jump”. The Q-Q plot from normal data is not an ideal line either.

Logarithmic substitution

Movie budgets: $n = 2,467$

Budget (\$m)	ln(budget)	Budget	ln(budget)
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Skewness	+1.618	-0.738	Q-Q	
	0.9007	0.9799		χ^2 35.81
Med	22.00	3.091		2.939

The logarithm significantly reduced the skewness to the right: χ^2 0.90 $\rightarrow$ 0.98.

Where the Normal Model Works, Where It Fails

Budget	z-score	Normal Model	Prediction	Actual	Percentile
\$10 million	-0.484	31.4%	28.7%	\$50 million	+0.741
76.5%				77.0%	
\$150 million	+1.577	94.3%	98.5%		

The model is excellent in the center. At \$50 million, the error is only 0.5 percentage points.

A common mistake is that it fails in the tail. At \$150 million, the model says 5.7% of the film is worth that much; in fact, it is 1.5%. The error is almost four times.

Why is this important? In risk management, insurance, and finance, decisions are made in the tail. Just because the normal model is good in the center, it cannot be trusted in the tail.

Exponential distribution

Exponential distribution: waiting time

):

$$f(t) = \lambda e^{-\lambda t}, \quad t \geq 0$$

$$F(t) = 1 - e^{-\lambda t}$$

$$E[\square] = \frac{1}{\lambda}, \quad SD(\square) = \frac{1}{\lambda}$$

Link to week 4. If the number of events is $\text{Pois}(\lambda)$, then the time between them is $\text{Exp}(\lambda)$. Two views of the same process.

Example (from week 4). $\lambda = 10.95$ horror movies per year.

$$E[\square] = \frac{1}{10.95} = 0.0913 \text{ years} = 33.3 \text{ days}$$

$$\text{Med} = \ln 2 \cdot \frac{1}{\lambda} = 0.0633 \text{ years} = 23.1 \text{ days}$$

$$P(\square > E[\square]) = e^{-1} = 0.368$$

Waiting time is longer than mean in 36.8% of cases.

Memoryless property

$$P(X > \tau + \Delta | X > \tau) = P(X > \Delta)$$

Proof.

$$P(X > \tau + \Delta) = e^{-\lambda(\tau + \Delta)} = e^{-\lambda\tau} e^{-\lambda\Delta} = P(X > \tau) P(X > \Delta)$$

Verification ($\lambda = 10.95$, $\Delta = 0.05$):

τ	$P(X > \tau + \Delta)$	$P(X > \tau) P(X > \Delta)$
0,05	0,578469	0,578469
0,10	0,578469	0,578469
0,20	0,578469	0,578469

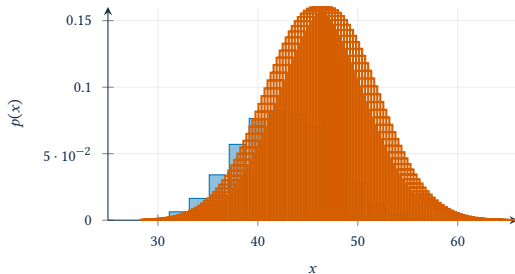
Meaning. No matter how long you wait, the remaining waiting time distribution does not change. The process "does not remember how much time has passed".

Common mistake Therefore, this model is often incorrect. The longer a person waits in line, the less patient they are. The longer a machine is in operation, the greater the chance of failure.

Memory loss is a convenience, not a law of nature.

Normal approximation of a binomial

Approximating the Binomial with the Normal



$\text{Bin}(100; 0.4528)$ and $\mathcal{N}(45.28; 4.982)$

Theorem If $np \geq 10$ and $n(1-p) \geq 10$:

$$\text{Bin}(n, p) \approx \mathcal{N}(np, np(1-p))$$

Common Mistake Continuity correction is required: we are replacing the discrete quantity with a continuous model.

$$P(a \leq X \leq b) \approx \Phi\left(\frac{b+0.5-\mu}{\sigma}\right) - \Phi\left(\frac{a-0.5-\mu}{\sigma}\right)$$

How important is the continuity correction?

	$n = 10$	$n = 30$	$n = 100$	
Condition $n\pi \geq 10, n(1 - \pi) \geq 10$	not met	met	met	Exact binomial
	0.798231	0.536146	0.311549	Normal + correction
	0.796051	0.536577	0.311877	Normal, uncorrected
	0.659302	0.417623	0.236626	
Error (with correction)	0.0022	0.0004	0.0003	Error (without correction)
	0.1389	0.1185	0.0749	

The correction reduced the error by a factor of 63 at $\pi = 10$ — even though the condition was not met.

Common Error The uncorrected error remains 0.075 even at $\pi = 100$ — increasing π does not compensate for the correction.

Summary

Common Student Mistakes and Misconceptions

Misconception Why it's wrong

" $P(x)$ is the probability of x ." $p(x)$ is the density; it can be greater than 1. Probability is just the area under the curve.

"A lot of data — so it's normal." 2,467 movie budgets Q-Q $\rho = 0.90$: not normal. The size of n does not make the shape normal.

"The skewness is close to zero — so it's normal." Uzbekistan and Kyrgyzstan have almost the same skewness (-0.65), but one is normal and the other is not. Look at the graph.

"The logarithm always restores normality." The budget skewness went from $+1.62$ to -0.74 — improved, but reversed.

"If the normal model works in the center, it works everywhere." At \$50 million, the error is 0.5 p.p.; at \$150 million, it's 4.2 p.p. The model breaks down in the tail — right where decisions are made.

" $P(x < y)$ and $P(x \leq y)$ are different." In the continuous case, there is no difference, because $P(x = y) = 0$. In the discrete case, there is a difference — don't confuse the two.

"The continuity correction is an optional thin spot." At $n = 10$, it reduces the error from 0.139 to 0.002. The normal approximation without the correction is almost useless.

Key Conclusions

1. Continuous probability is a surface. A point probability is zero.
2. $z = (x - \mu) / \sigma$ reduces any normal distribution to a single standard scale.
3. Normality is an assumption, testable. A Q-Q plot shows more than just one number.

4. Substitution (e.g. $\ln$) helps, but does not guarantee perfection.
5. The normal model is reliable in the center, not in the tails.
6. If you replace discrete with continuous, add a continuity correction.

Next Week Week 6 — Joint Distributions, Covariance, and Correlation. How two random variables behave together; portfolio variance; Anscombe Quartet. Control Work 2 Week 6 (Weeks 4-6).

This week's assignments

Assignment Content Due Date

Required Reading Newbold Chapter 5; McClave Chapter 5; Heumann Chapter 8 Before Lecture 1 Lab W05-lab — density, $\hat{\mu}$ -estimate, Q-Q graph, normal approximation In the lab Homework W05-homework — 6 problems, 40 points Week 6, Monday Problem set 3 MT3 due this week Problem set 4 MT4 due: Week 6 Project Step 5 Data cleaning: missing values, outliers Week 6, Friday

SI assignment Find SI's errors about normality Week 6, Friday

Common error in Week 6 Control work 2 — material from Weeks 4–6 (random quantities, discrete and continuous distributions).

1. Newbold, P., Carlson, W. & Thorne, B. Statistics for Business and Economics. Pearson. Chapter 5 — Continuous Random Variables and the Normal Distribution.
2. McClave, J. T. & Sincich, T. (2018). Statistics, 13th ed. Pearson. Chapter 5.
3. Heumann, C., Schomaker, M. & Shalabh (2016). Introduction to Statistics and Data Analysis. Springer. Chapter 8.
4. Çetinkaya-Rundel, M. & Hardin, J. (2024). Introduction to Modern Statistics, 2nd ed. OpenIntro — Normal Distribution and Q-Q Graphs.
5. DeGroot, M. & Schervish, M. (2012). Probability and Statistics, 4th ed. Chapter 5 (in-depth).
6. Limpert, E., Stahel, W. & Abbt, M. (2001). Log-normal distributions across the sciences. *BioScience*, 51(5), 341–352.
7. Interactive: Seeing Theory, Brown University, "Probability Distributions" — seeing-theory.brown.edu
8. Video: 3Blue1Brown, "Why π is in the normal distribution"; StatQuest, "The Normal Distribution, Clearly Explained."